

Omninet_AI_Marketing_Requirements_Document_v1.0

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1. Scope

This document specifies the requirements for a next-generation family of Internet kiosks called **Omninet AI**. These kiosks shall provide users with secure, high-speed, and intelligent Internet access in public places. The system will support modern payment methods and leverage artificial intelligence to enhance user experience, security, and operational efficiency.

1.1 Terms, Acronyms, and Abbreviations

For the purposes of this project, the following updated abbreviations apply:

AI Agent	An intelligent software entity for automation, support, and analytics.
API	Application Programming Interface
CC	Credit Card (Visa, MasterCard, American Express, Discover)
Cloud	Cloud-based infrastructure for hosting all Omninet services.
DL	Deep Learning model
Fiber	Fiber-optic Internet connection, minimum 100 MBPS.
5G/WiFi6	Primary high-speed connectivity for kiosks (cellular/wireless).
Kiosk	The free-standing Omninet AI access point.
LLM	Large Language Model (for AI-powered support and interaction)
MBPS	Megabits per second

MobilePay	Digital wallet payment (Apple Pay, Google Pay, etc.).
NFC	Near Field Communication (for contactless payments).
OS	Operating System (e.g., Windows 11 IoT, Ubuntu Linux, Android).
SSO	Single Sign-On
URL	Universal Resource Locator

1.2 Applicable Documents

[1] See the Omninet AI System Architecture Document for cloud and edge deployment designs.

[2] See the Omninet AI UX/UI Prototype Document for the kiosk and support center dashboards.

2. Required Release Date

The first 1,000 Omninet AI kiosks shall be operational, accepting payments and providing intelligent Internet access, by Q3 2026.

3. Description of Requirements

3.1 General Technical Requirements

Omninet AI shall provide users in airports, malls, hotels, and other public places with secure, personalized, and high-speed Internet access.

The Omninet AI system comprises three primary subsystems:

1. **The Kiosk (Edge Device):** The physical unit users interact with.
2. **The Omninet Cloud (Core Services):** A centralized platform for management, AI processing, and data analytics.
3. **The AI Support Center (Intelligent Interface):** A dashboard for human agents, augmented by AI agents, to monitor and manage the network.

3.1.1 Welcome

Between sessions, each Omninet AI kiosk shall display an inviting, dynamic welcome message, potentially personalized based on location and time of day (see screen prototype K.1).

3.1.2 Payment

Once a user initiates a session, the kiosk shall offer flexible payment options.

The system shall use AI to suggest optimal session lengths based on location and historical usage patterns.

The system accepts the following forms of payment:

- **Traditional:** Bills and coins (see screen prototype K.3).
- **Modern Cards:** Credit/Debit cards with chip and NFC support (see screen prototype K.4, K.5, K.7).

- **Digital & Mobile:** Mobile wallets (Apple Pay, Google Pay), and QR code payments (e.g., WeChat Pay, Venmo) (see screen prototype K.8).
- **AI-Powered Promotions:** The system may offer promotional time in exchange for viewing an ad or completing a quick survey.

Time is sold in flexible increments, with a minimum of 1 minute and a maximum of 2 hours. An AI agent will predictively prompt the user to purchase more time based on their browsing activity and remaining time (see screen prototype K.9).

3.1.3 Internet Browser & AI Assistant

At the Welcome screen, each Omninet AI kiosk shall launch a secure, modern browser (e.g., latest Chromium, Firefox). The browser will be integrated with an **Omninet AI Assistant**—a conversational AI agent that can:

- Answer questions about the kiosk's features.
- Help find information or navigate to websites.
- Translate on-screen text or web pages in real-time.
- Summarize articles or content for the user.

3.1.4 Performance

On all kiosks, users shall have a minimum connection speed of 100 MBPS, provided via 5G, WiFi6, or Fiber backhaul, ensuring a seamless modern web experience.

3.1.5 Localization

Each Omninet AI kiosk shall detect the most likely user language based on location or offer a selection at the Welcome screen. The integrated AI Assistant shall support real-time conversational translation, allowing users to interact in their preferred language.

3.1.6 Content Control & Security

Omninet AI shall employ a multi-layered security approach:

- **AI-Powered Content Filtering:** Instead of simple site blocking, an on-device AI agent will analyze content in real-time to block or blur pornographic, violent, or otherwise objectionable material, while allowing access to informative content like health sites.
- **Advanced Threat Protection:** Each kiosk shall run continuous, AI-driven security scans to detect and prevent zero-day exploits, ransomware, and other malicious activities, protecting both the user and the device.

3.1.7 Session Termination

Users may terminate sessions by:

- Logging out (no refund for unused time).
- Allowing purchased time to expire.

3.1.8 Confidentiality & Privacy

To ensure complete user privacy, upon session termination, each kiosk shall:

- Execute a secure wipe of all session data (cache, cookies, history, downloaded files) from the local device.
- Restart the browser in a fresh, sandboxed environment at the Welcome screen.
- The AI Assistant's conversation log for the session is permanently deleted.

3.2 Administration (via Omninet Cloud & AI Support Center)

3.2.1 Software & AI Model Updates

Software and security updates shall be managed centrally from the Omninet Cloud.

- At 2:00 AM local time, each kiosk shall check for and apply updates, including:
 - OS and browser patches.
 - Device drivers.
 - **AI Agent behavioral models and content filtering datasets.**
 - Updated payment rate tables.
 - Dynamic threat intelligence feeds.

- The update process is managed by an **AI Orchestrator Agent** in the cloud. If the update server is overloaded, the orchestrator assigns each kiosk a randomized retry window (5 to 30 minutes) to distribute load.
- Critical security updates or AI model patches can be pushed immediately by the AI Support Center.

3.2.2 View Kiosks (AI-Powered Monitoring)

The AI Support Center dashboard shall provide a real-time, intelligent view of all kiosks. For each kiosk, agents can see:

- Current OS and software versions.
- AI Agent status and health.
- Uptime and performance metrics.
- Number of crashes or hard failures.
- **AI-Predicted Failure Risk:** An AI analytics agent highlights kiosks at high risk of imminent failure, enabling proactive maintenance.
- Kiosks report their status to the cloud every 15 minutes.
- If a kiosk is offline or malfunctioning, it is highlighted in red at the top of the list. An AI agent will have already attempted a remote diagnostic and reboot cycle before alerting a human agent.

3.2.3 View Users (with AI Oversight)

For active sessions, the AI Support Center shall provide agents with relevant information, with strict privacy controls and audit trails. An **AI Monitoring Agent** flags anomalies for human review. Accessible information includes:

- **Anonymized Browsing Patterns:** The AI agent analyzes URLs visited for security threats, but a human agent can only view the full history if a security or fraud alert is triggered.
- Last four digits of the payment card (for verification).
- Amount paid and time purchased/remaining.
- **AI-Generated Risk Score:** A real-time score indicating the likelihood of fraudulent or inappropriate use.

3.2.4 Modify User

Human agents, with supervisory approval for large adjustments, can modify a user's session (e.g., adding time for compensation). An **AI Billing Agent** automatically handles common modifications, such as refunding time lost due to a temporary connectivity issue at the kiosk.

3.2.5 Terminate User & AI-Enhanced Intervention

If an **AI Monitoring Agent** detects highly probable illegal activity (e.g., credit card fraud patterns, network intrusion attempts, confirmed access to known

illegal material), it can automatically terminate the session and lock the kiosk, immediately alerting a human agent and local authorities if necessary.

For inappropriate but non-critical behavior flagged by the AI, a human agent can review and manually terminate the session. In both cases:

- The user is presented with a clear message explaining the reason for termination and the amount of any refund.
- If a refund is due (e.g., for a mistaken flag), the **AI Billing Agent** processes it automatically.